

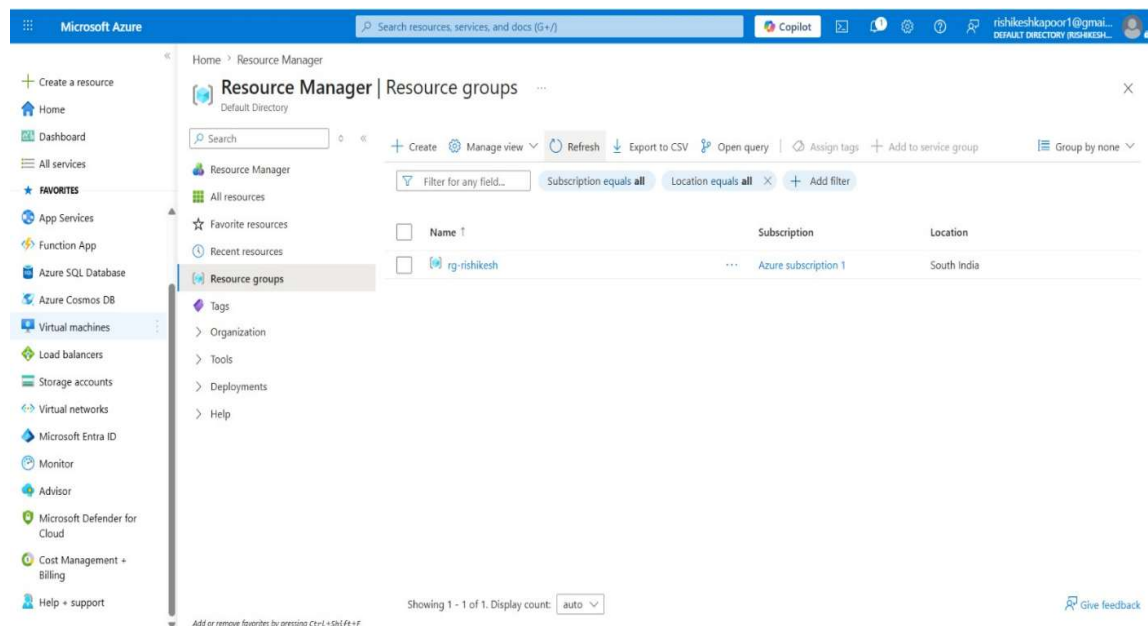
WEEK 4 ASSIGNMENT

INTRODUCTION

This assignment demonstrates the implementation of Azure Blob Storage and Azure Data Factory for copying data from one storage container to another. It also includes IAM role assignment and the use of the Get Metadata activity for retrieving file information before data movement.

TASK 1 – Create Resource Group

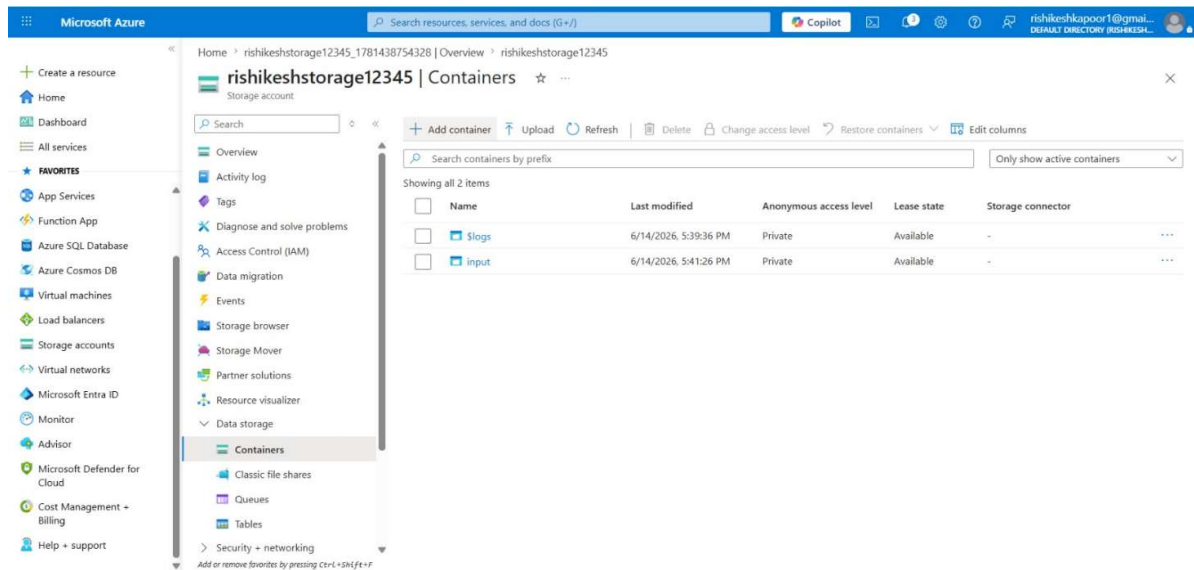
- Resource Group Name: rg-rishikesh
- Region: South India



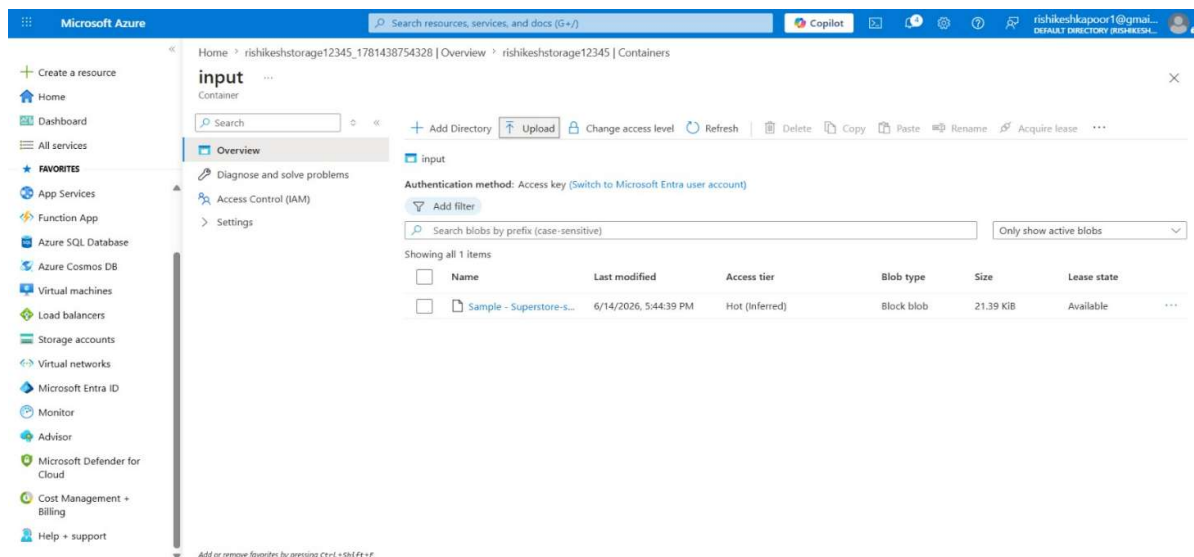
Task 2 – Create Storage Account

Configuration

- Storage Account Name: rishikeshstorage12345
- Performance: Standard
- Redundancy: Locally Redundant Storage (LRS)
- Region: South India



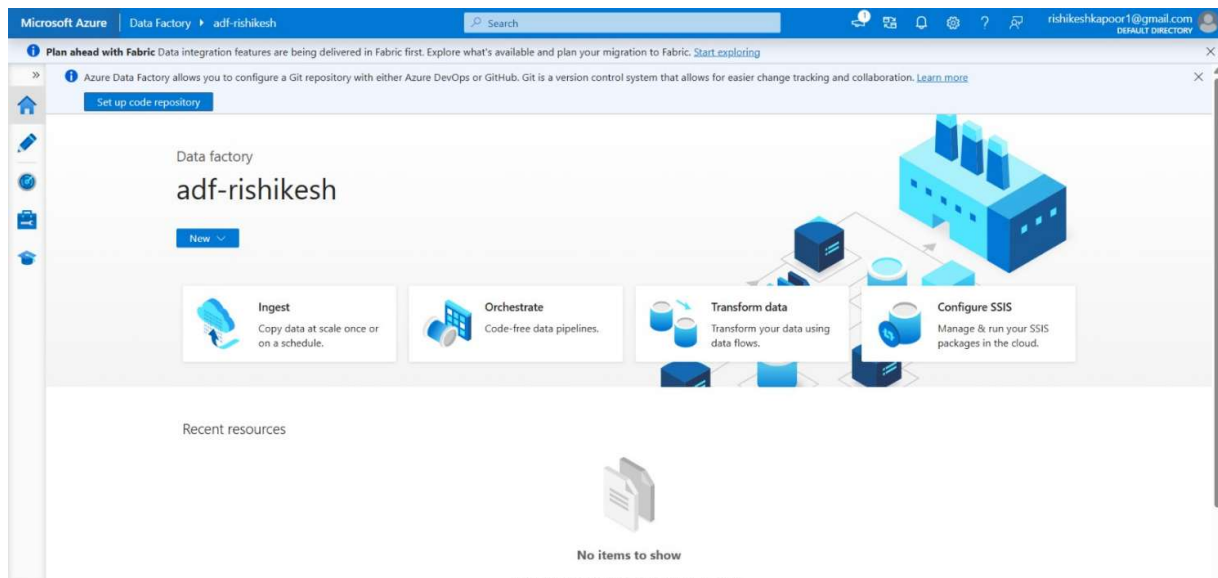
Upload Source CSV File



TASK 3 – ADF Basics

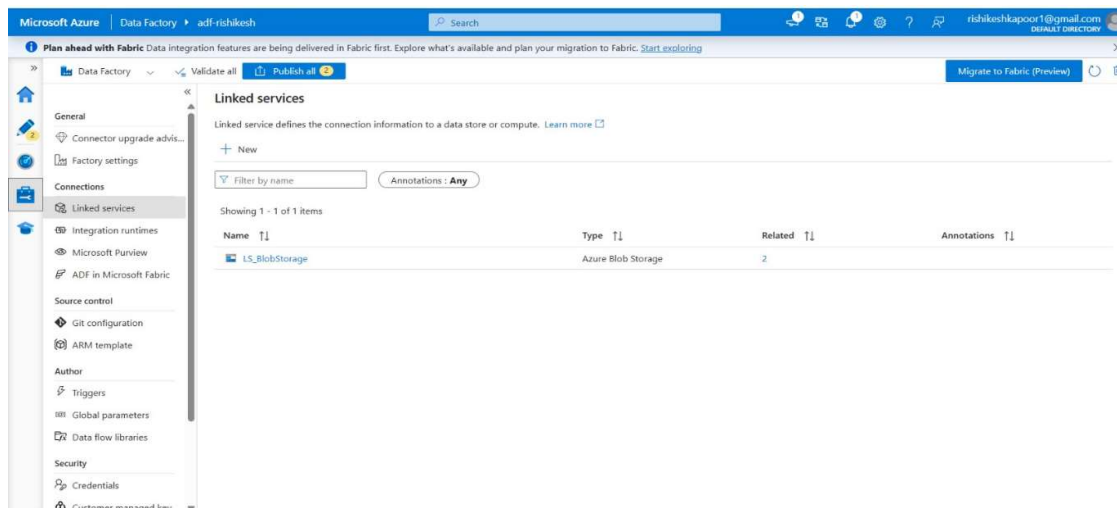
• Creation of Azure Data Factory

Azure Data Factory is a cloud-based data integration service provided by Microsoft Azure. It enables users to create, schedule, and manage data pipelines that move and transform data across different storage systems and services.



- **Create Linked Service**

A **Linked Service** acts as a connection between Azure Data Factory and external data sources. In this assignment, a Linked Service named **LS_BlobStorage** was created to establish a connection between Azure Data Factory and the Azure Blob Storage account. This connection allows the pipeline to securely access files stored in the storage account.

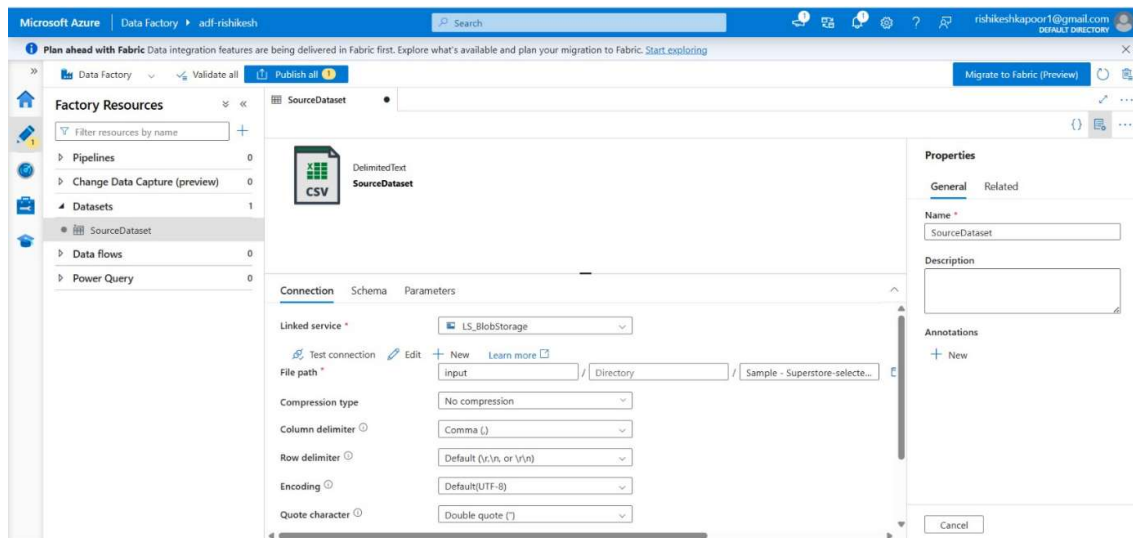


- **Create Source and Destination Datasets**

A Dataset represents the structure and location of data that will be used in a pipeline.

- Name: SourceDataset
- Container: input
- File: Sample - Superstore.csv

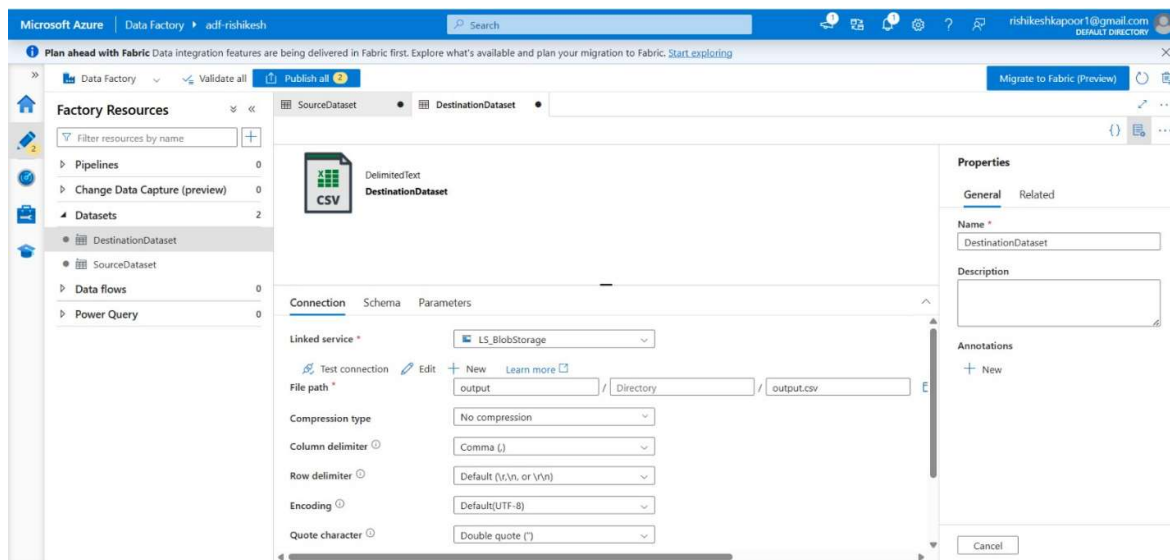
This dataset represents the input file that will be read by the pipeline.



Destination Dataset

- Name: **DestinationDataset**
- Container: **output**
- File: **output.csv**

This dataset represents the destination where the copied data will be stored.



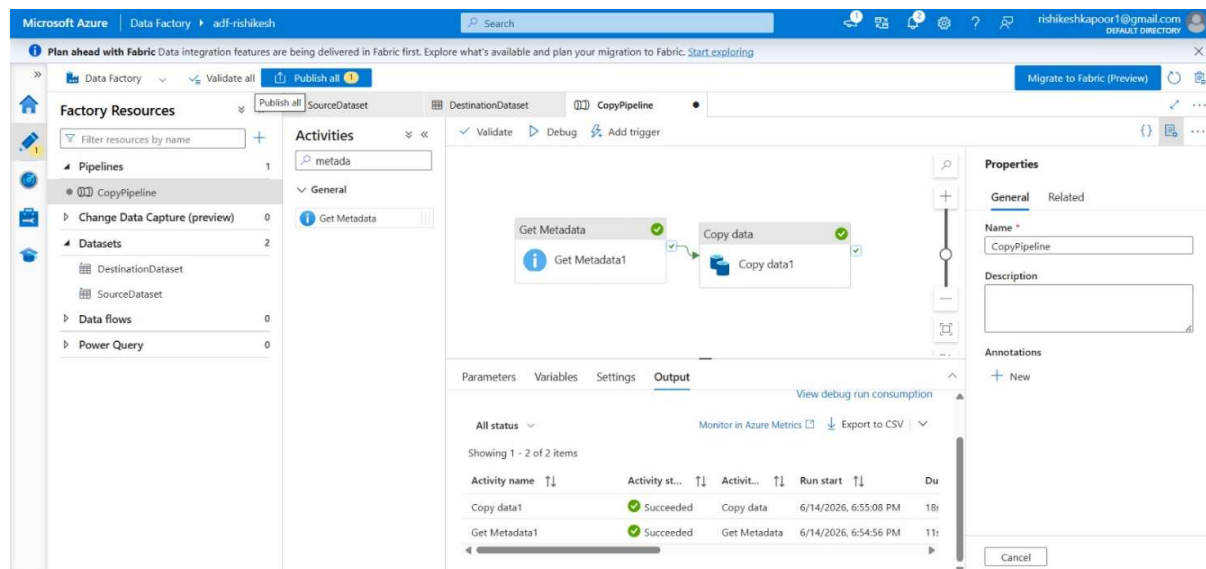
Use Get Metadata Activity

The **Get Metadata** activity is used to retrieve metadata information about a file or dataset before performing further operations.

In this assignment, the activity was configured with the **SourceDataset** and the **Exists** property was selected to verify whether the source file is available before executing the copy operation.

Purpose of Get Metadata

- Check whether the source file exists.
- Validate the availability of data before processing.
- Improve pipeline reliability by avoiding execution errors.



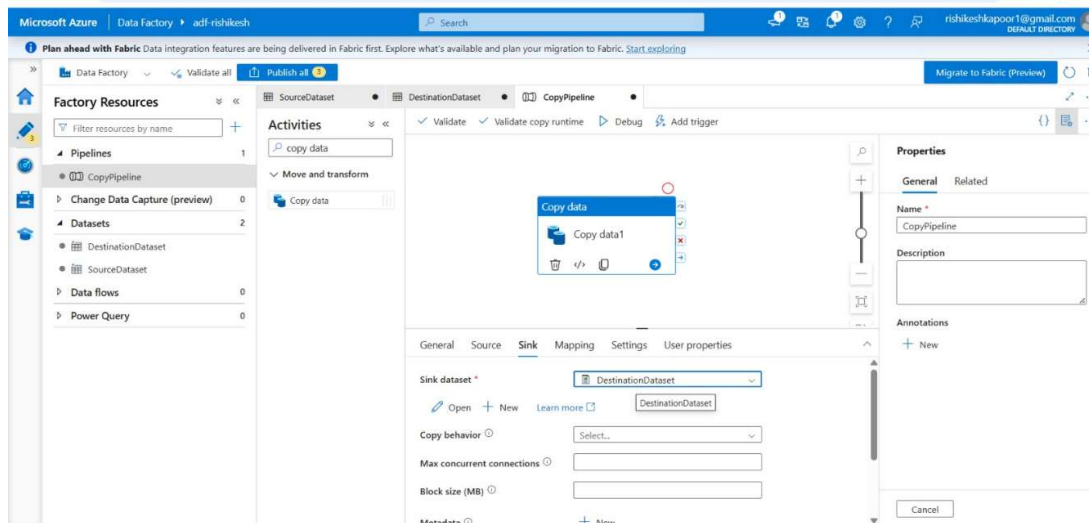
Task 4: Pipeline Development

Objective

The objective of this task is to create an Azure Data Factory pipeline that copies data from a source dataset to a destination dataset using the **Copy Data** activity.

Pipeline Development

A pipeline named **CopyPipeline** was created in Azure Data Factory. The **Copy Data** activity was configured to read data from the **SourceDataset** (stored in the input container) and copy it to the **DestinationDataset** (stored in the output container).



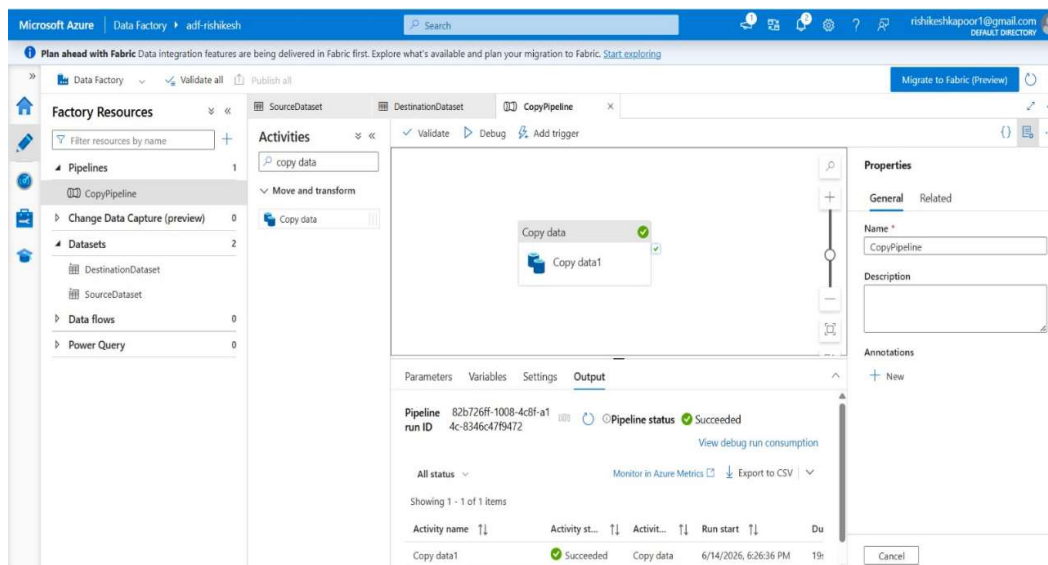
Task 5: Pipeline Execution

Objective

The objective of this task is to execute the Azure Data Factory pipeline and verify that the data transfer process completes successfully.

Pipeline Execution

The pipeline was executed using the **Debug** option in Azure Data Factory. During execution, the pipeline copied the data from the **SourceDataset** in the input container to the **DestinationDataset** in the output container.



Task 6: IAM Roles

Objective

The objective of this task is to manage access permissions by assigning appropriate IAM (Identity and Access Management) roles to users and Azure Data Factory.

Role Assignment

The **Reader** role was assigned to provide read-only access to the storage resources. Additionally, the **Storage Blob Data Contributor** role was assigned to Azure Data Factory to allow it to read and write data in Azure Blob Storage, enabling successful pipeline execution.

